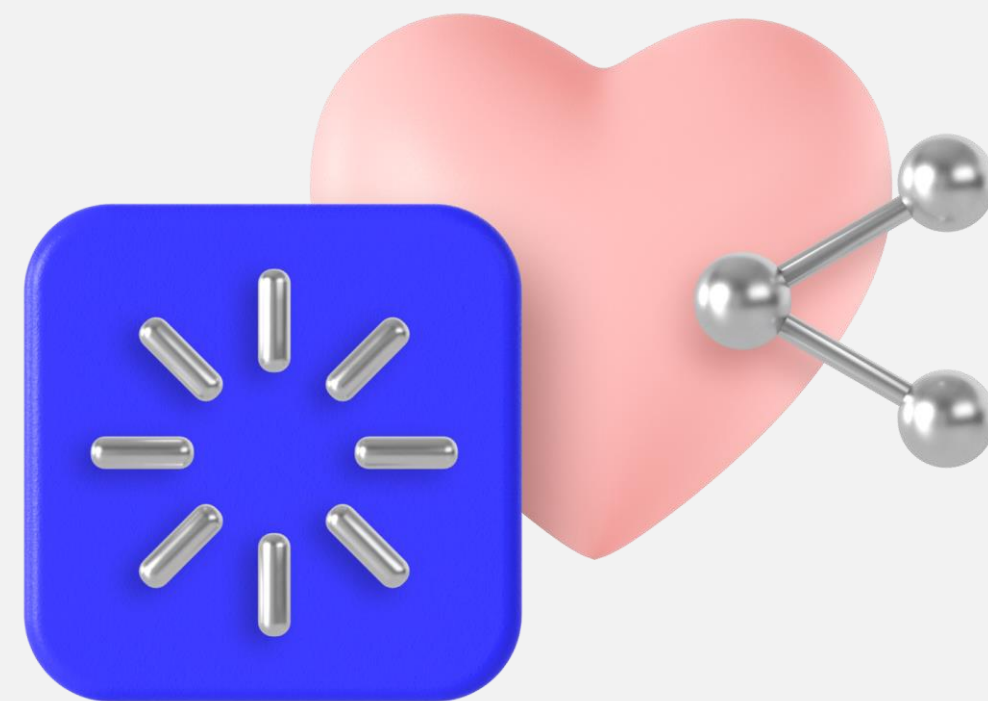


Объектные хранилища

Цели

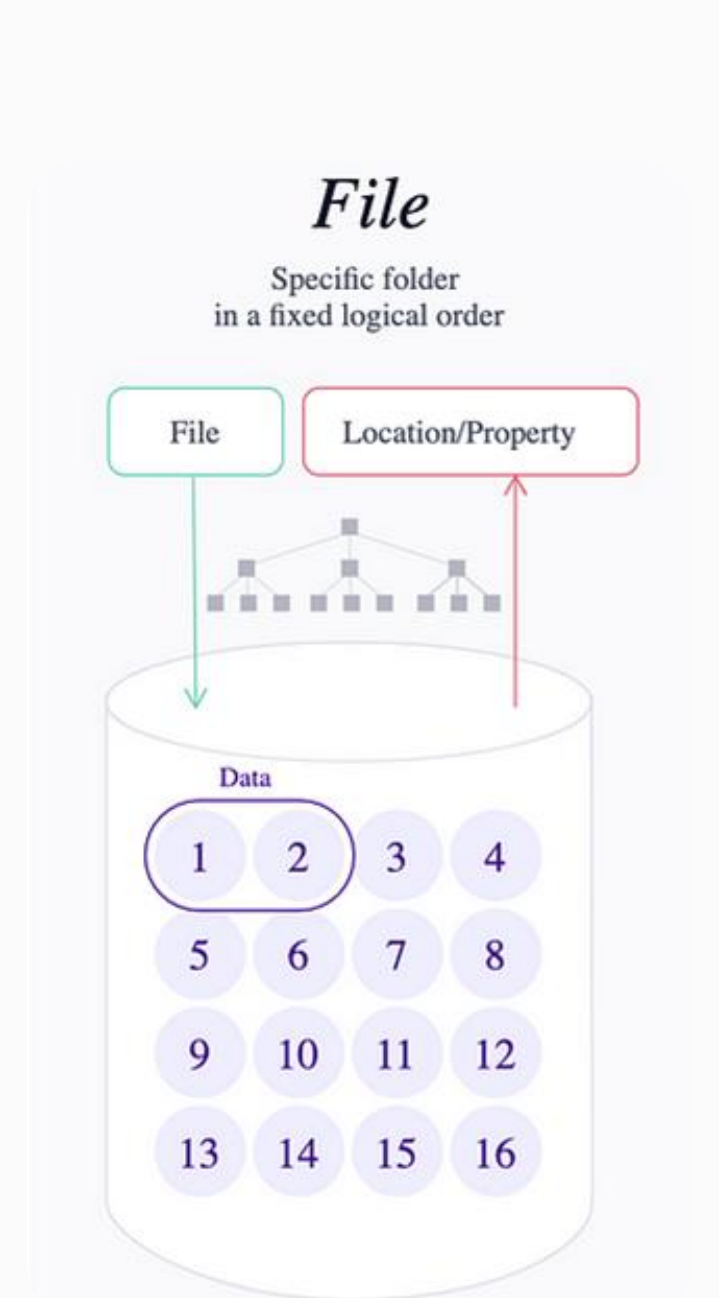
- ✓ Узнать, что такое объектное хранилище
- ✓ Познакомиться с преимуществами объектных хранилищ

File vs Object storage



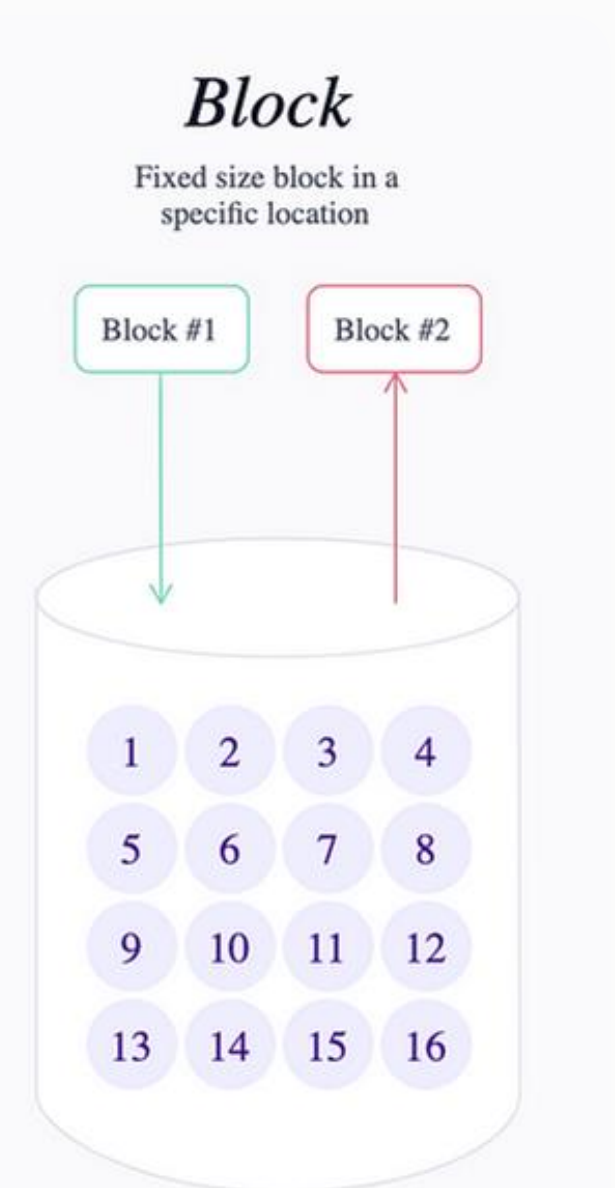
File storage

- 1 hierarchical structure
- 2 without breaking
- 3 4 KB blocks



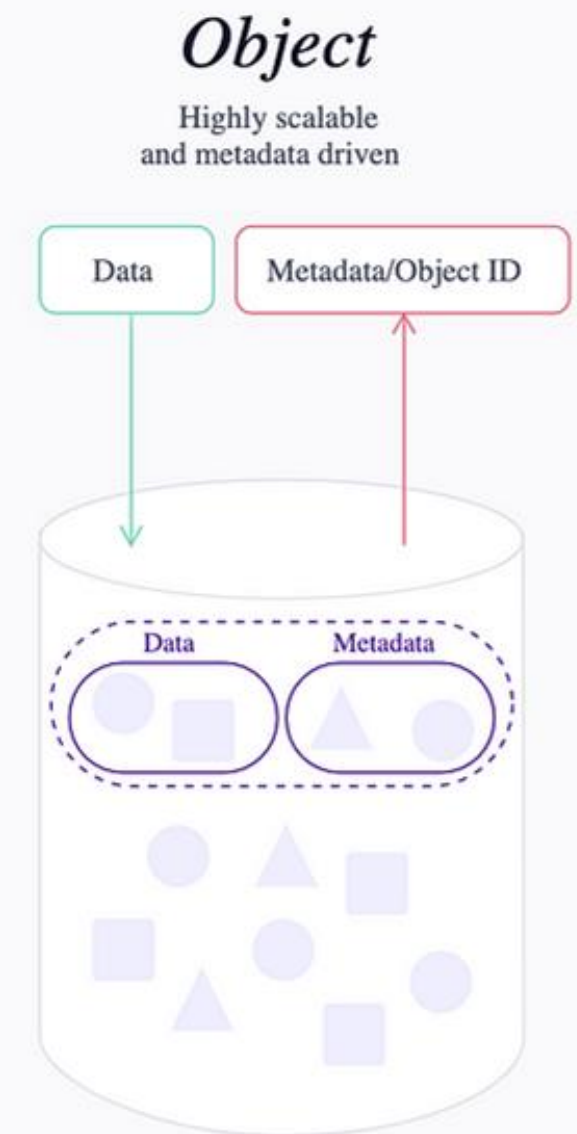
Block storage

- 1 Data is stored in blocks
- 2 Block has unique identifier
- 3 Storage-area network



Object storage

- 1 Data is stored in blocks
- 2 Block has unique identifier
- 3 Storage-area network



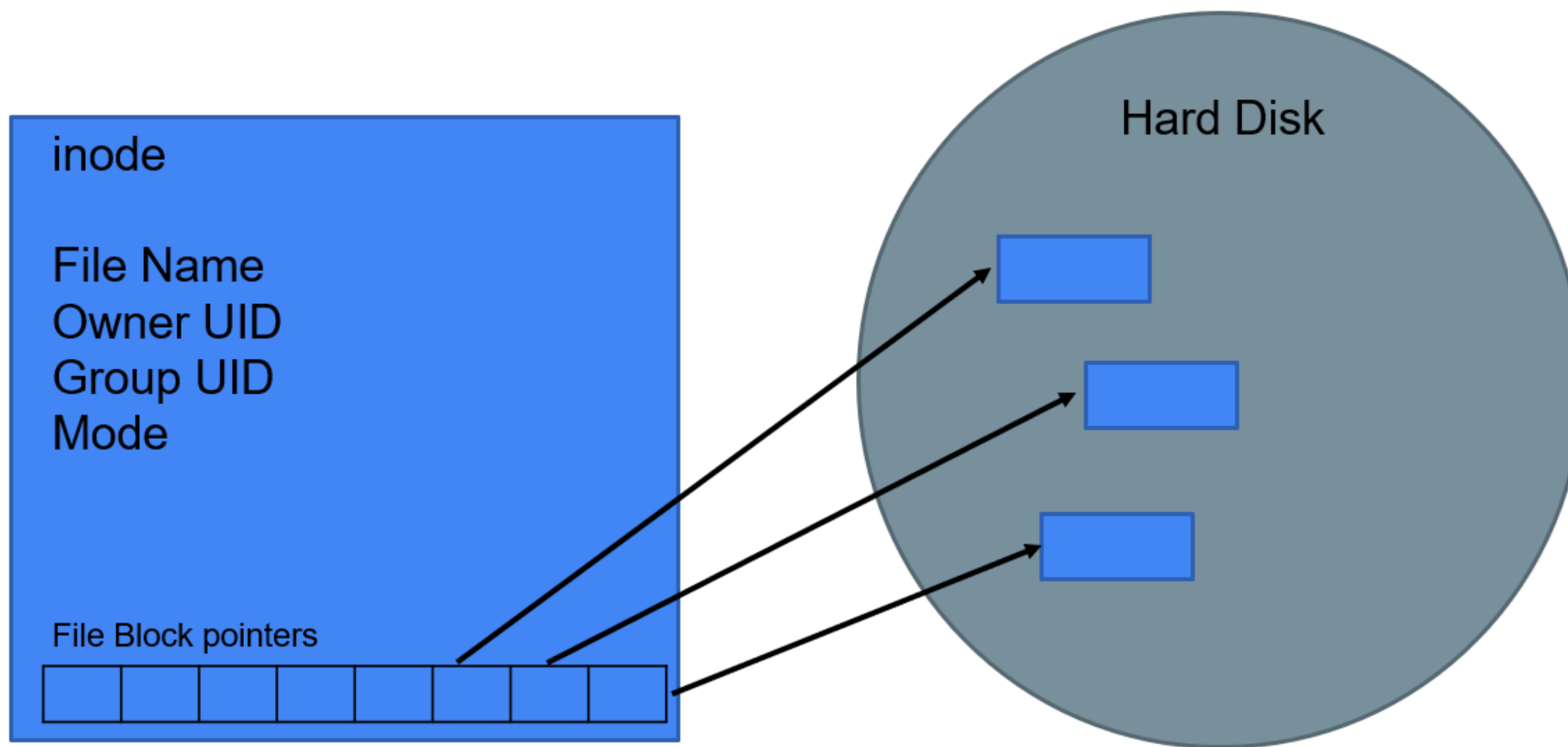
Некоторые понятия необходимые для object storage

- 1 Bucket — logical container
- 2 Object — data (byte seq) and meta data(key value)
- 3 Versioning — feature
- 4 URI — Uniform Resource Identifier

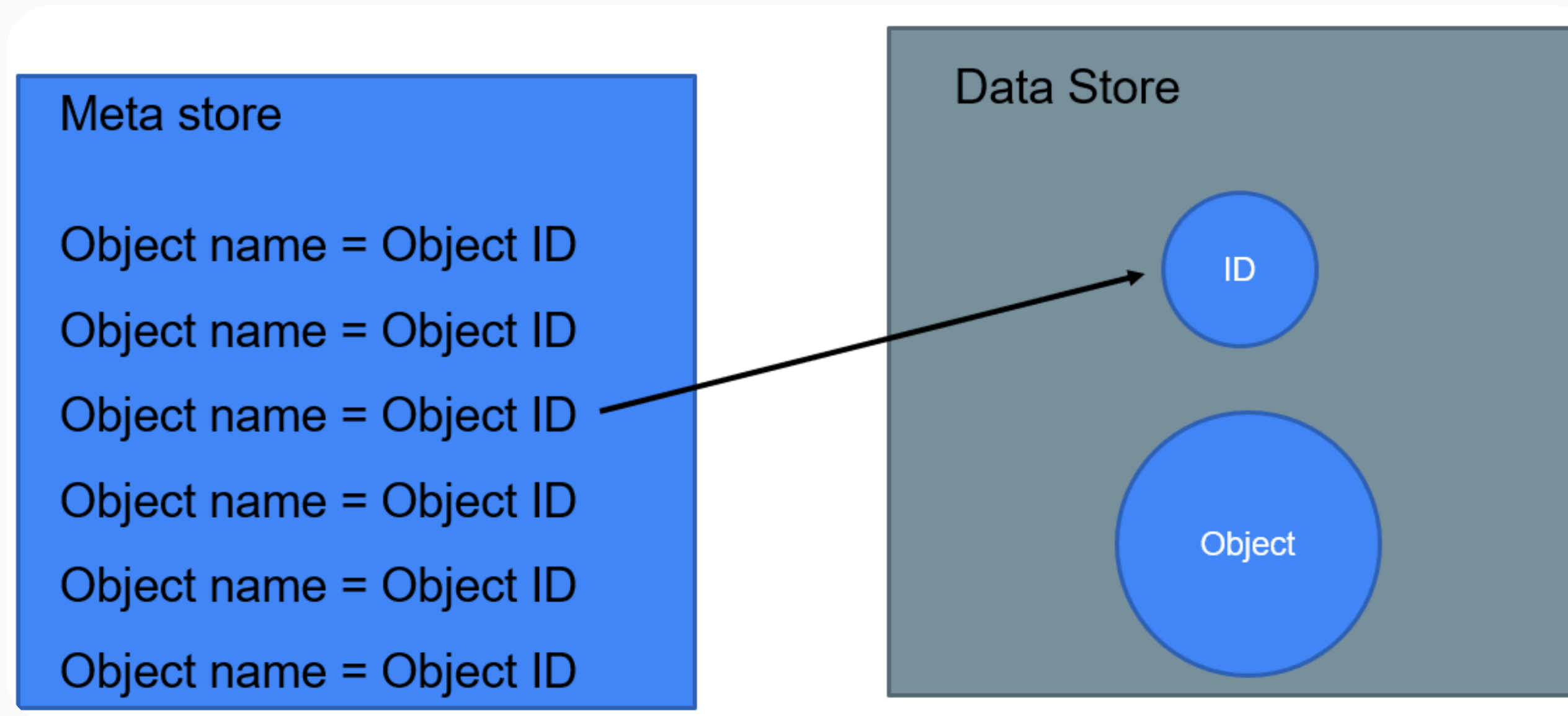
Свойства

- ❶ Object immutability — object are immutable
- ❷ Key — Value store — object URI (key), object payload (Value)
- ❸ Write Once, Read Many Times — 95 % READ
- ❹ Support large and small objects

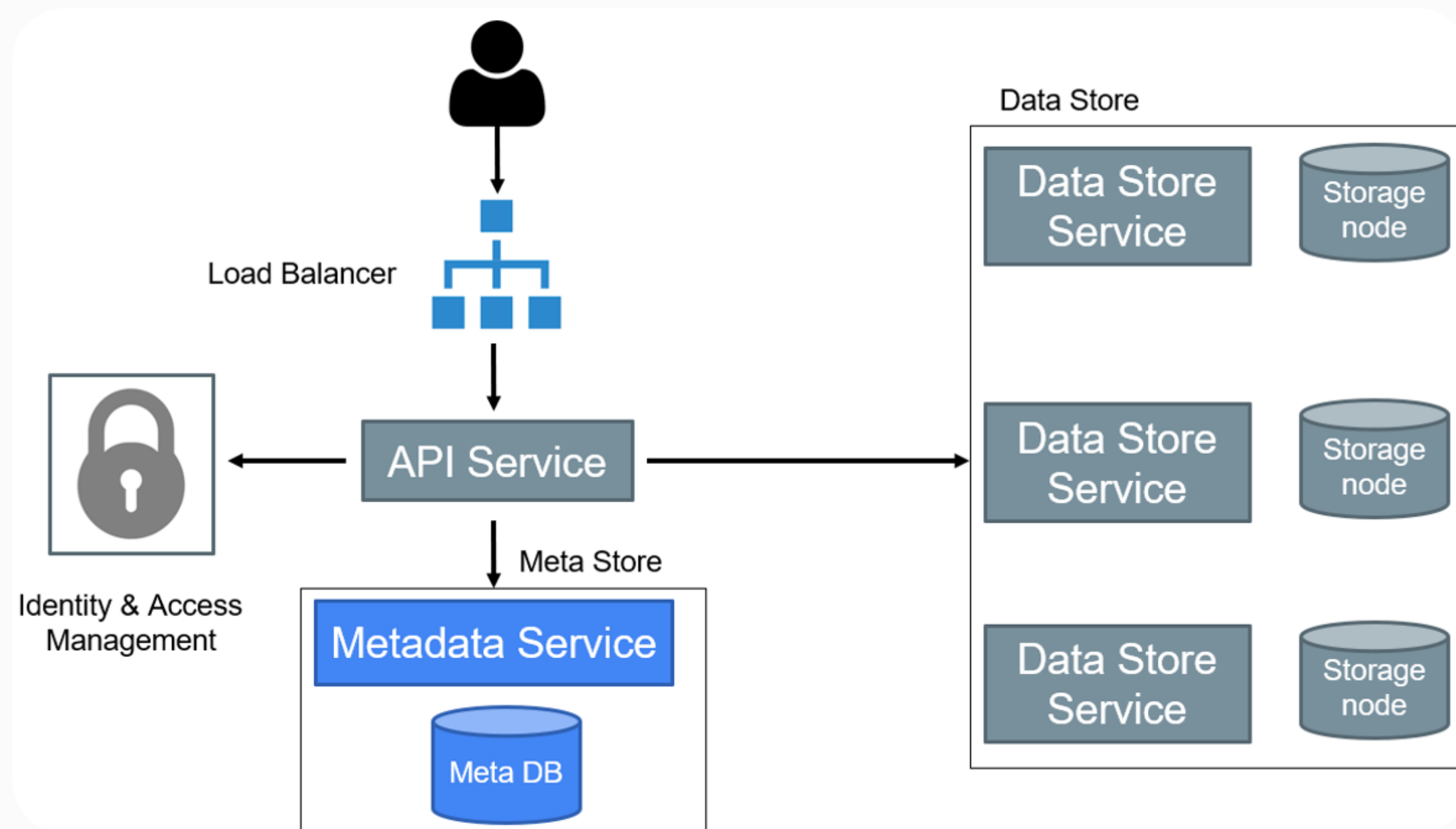
Unix file system



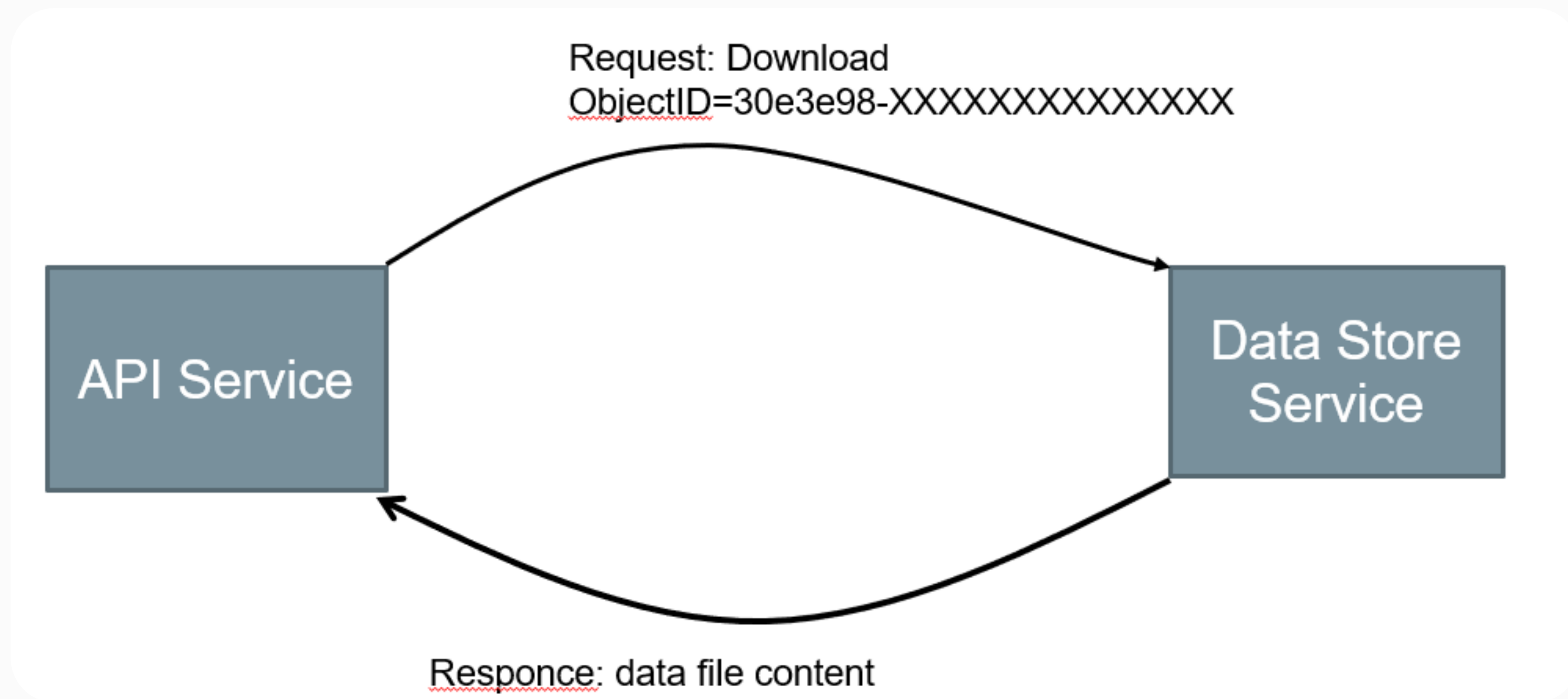
Object storage



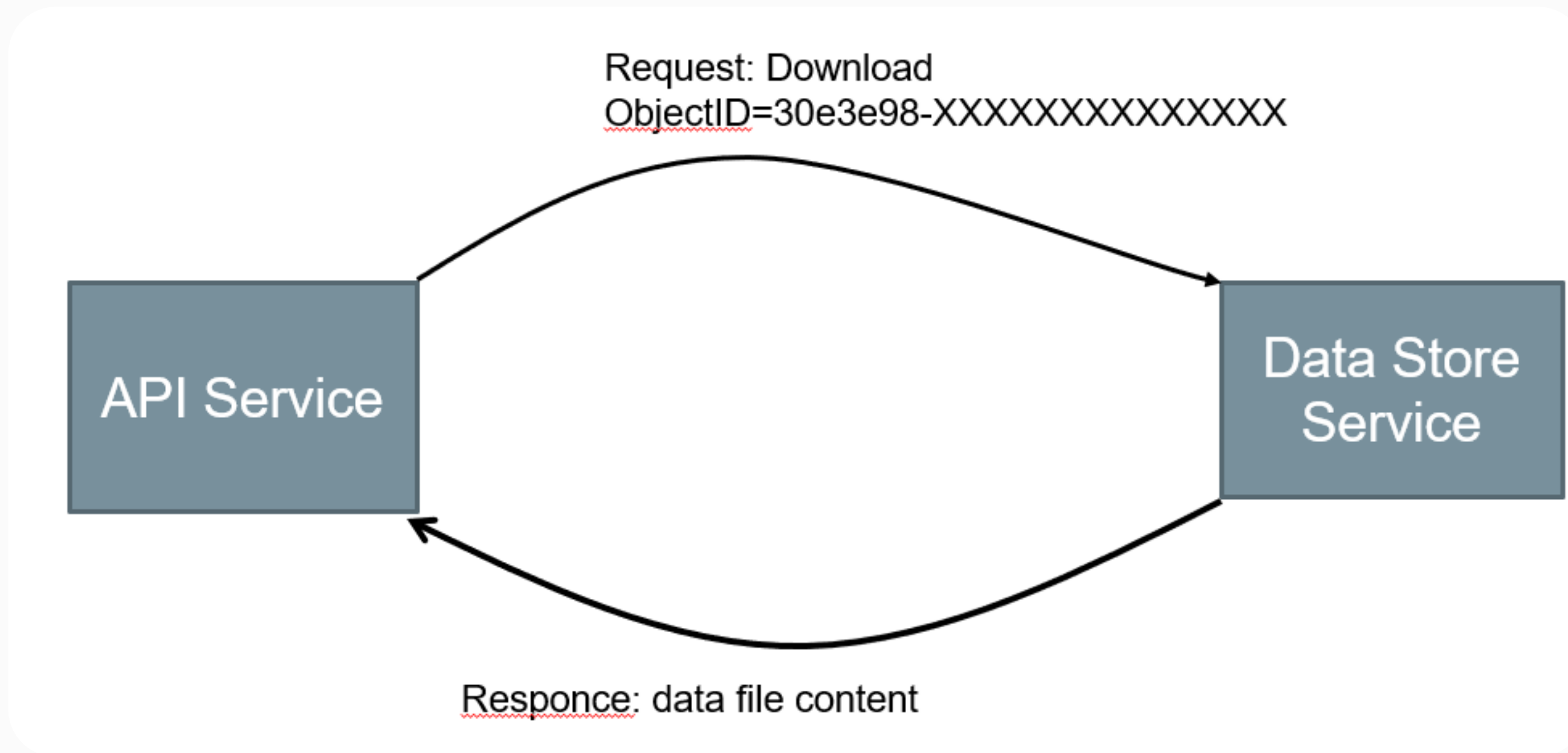
Архитектура



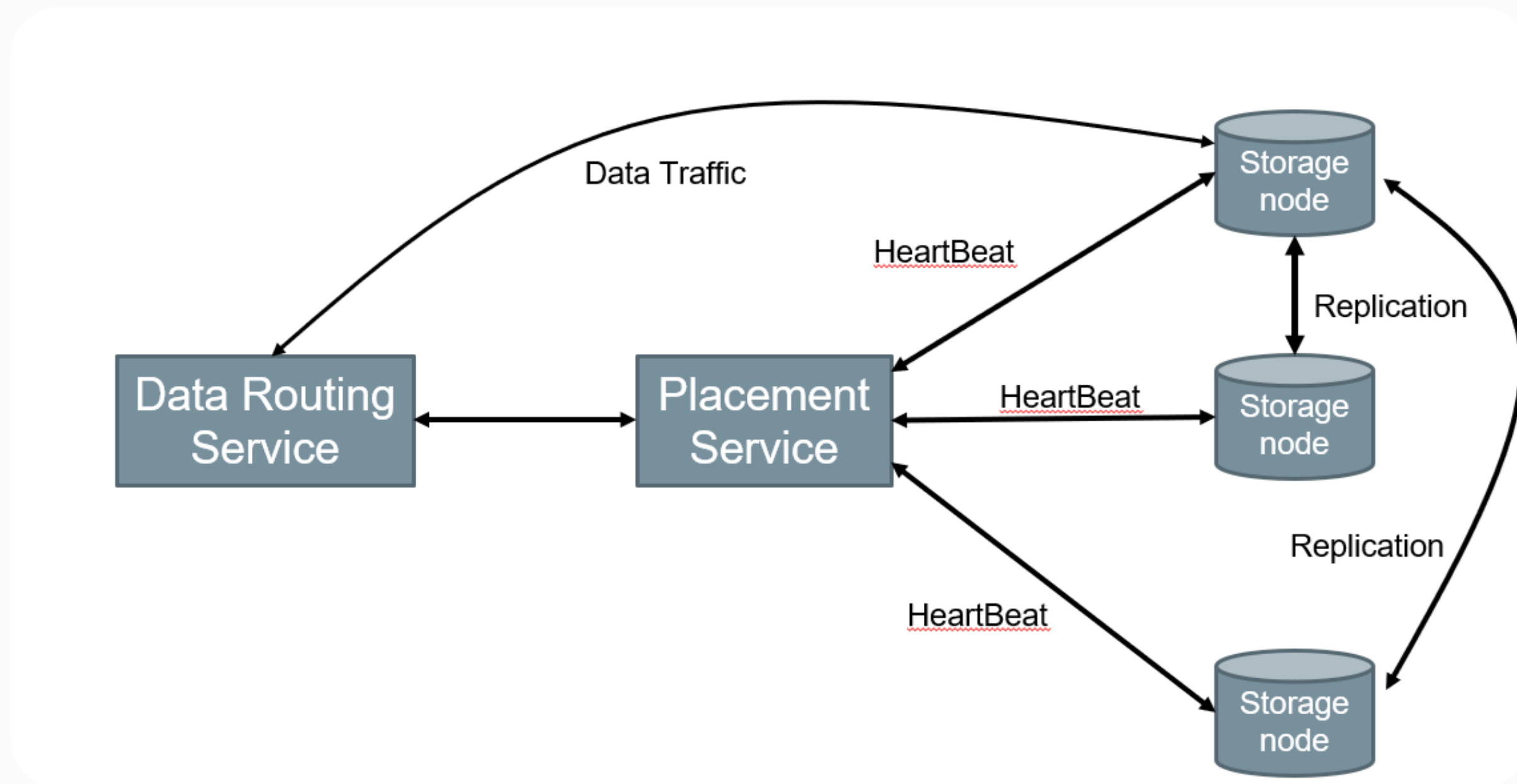
Запись



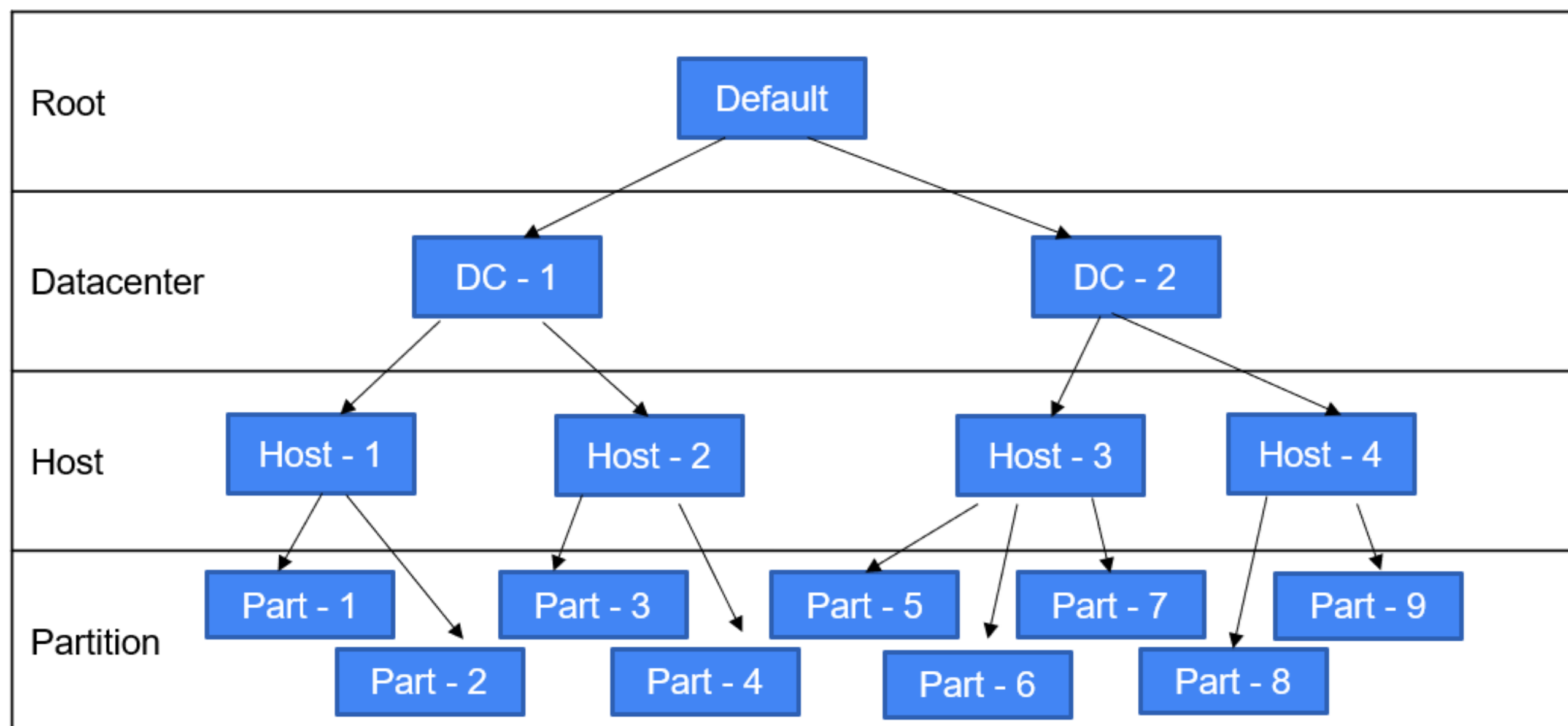
Чтение



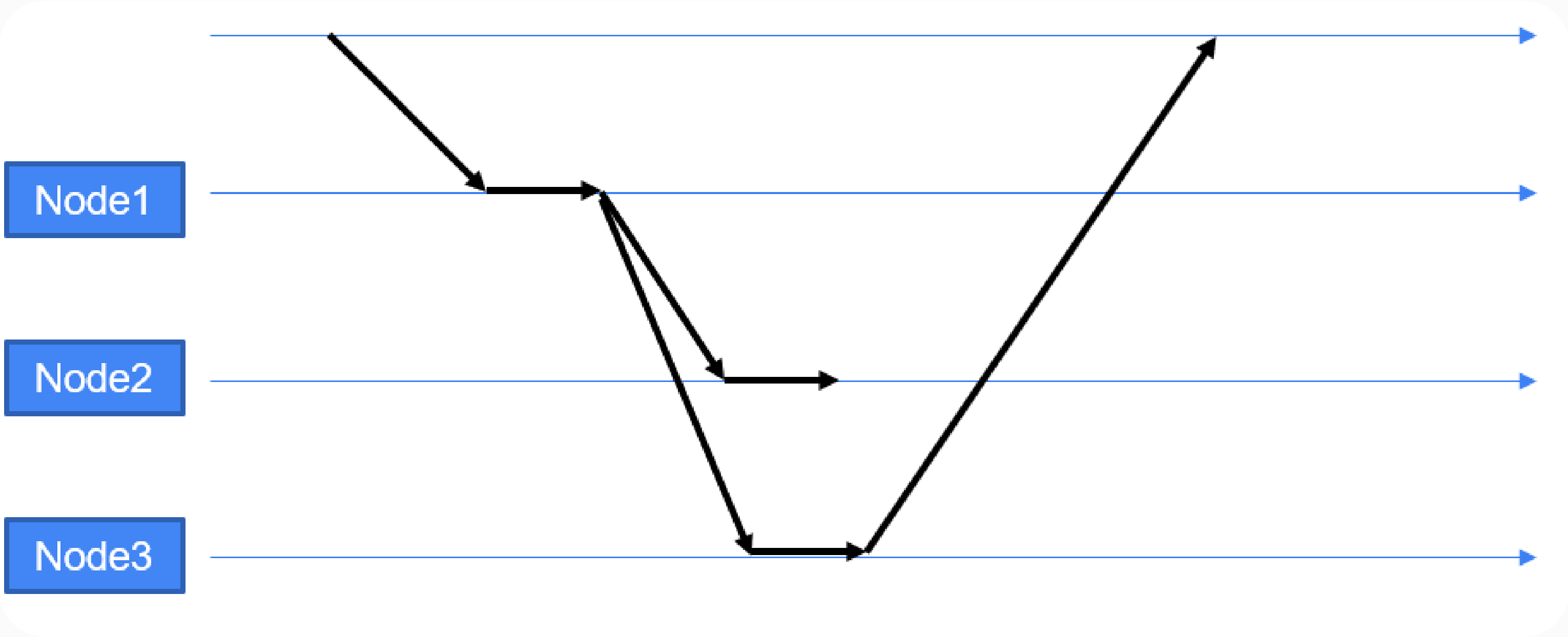
Data Store



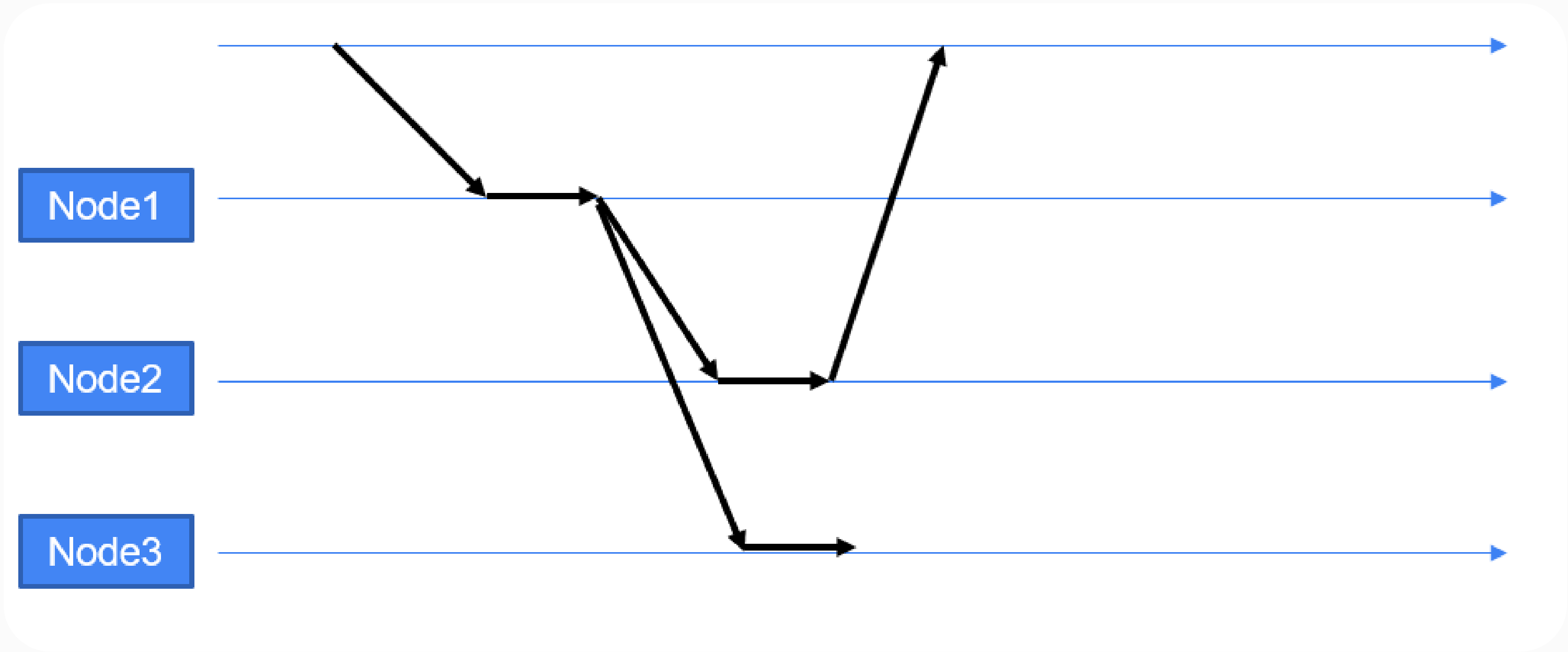
Карта



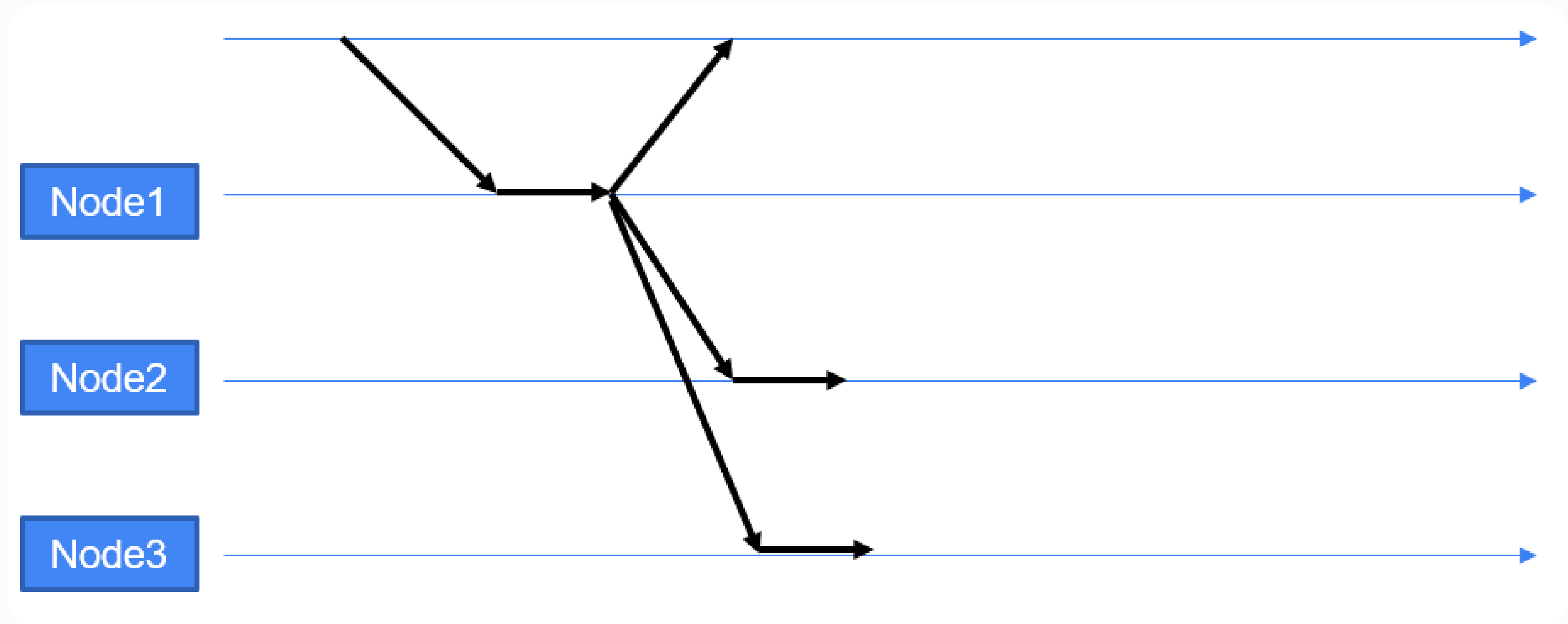
Best consistency latency



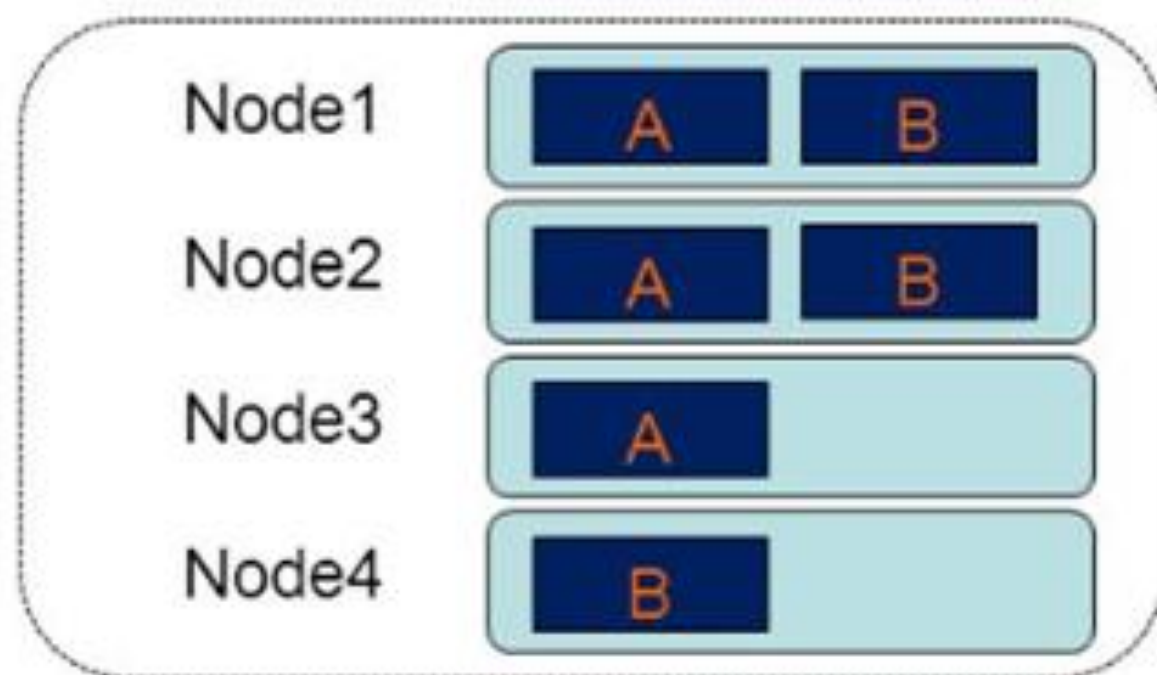
Medium consistency latency



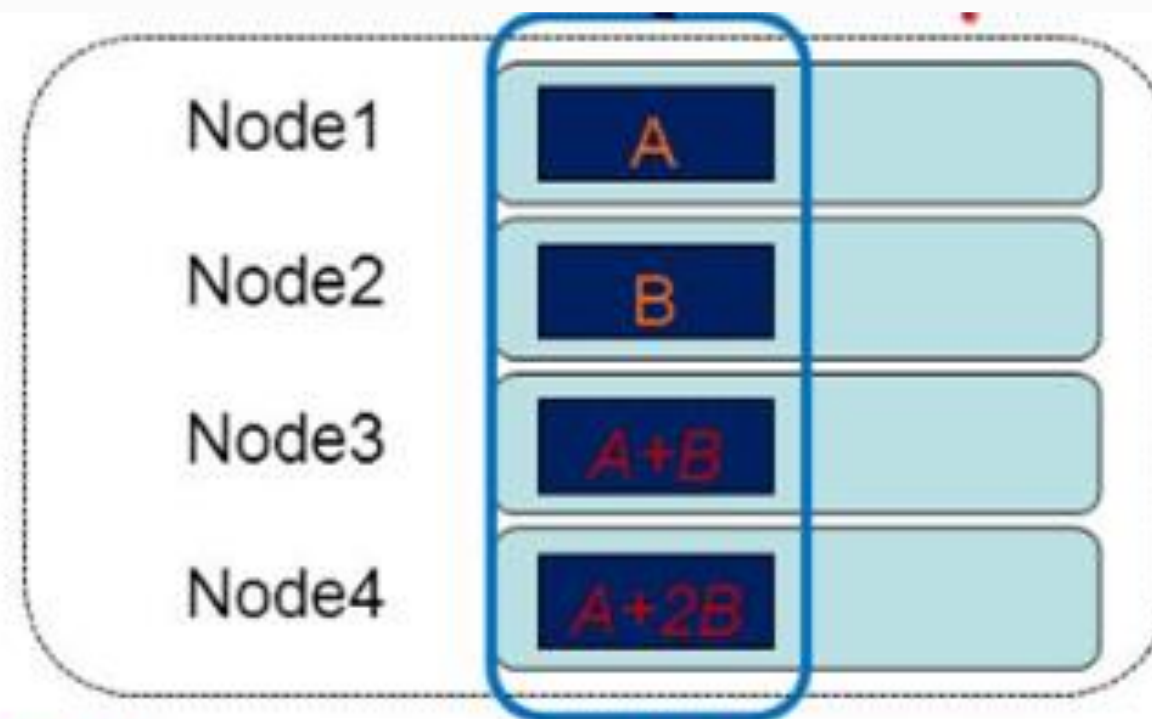
Medium consistency latency



Replication vs Ensure coding

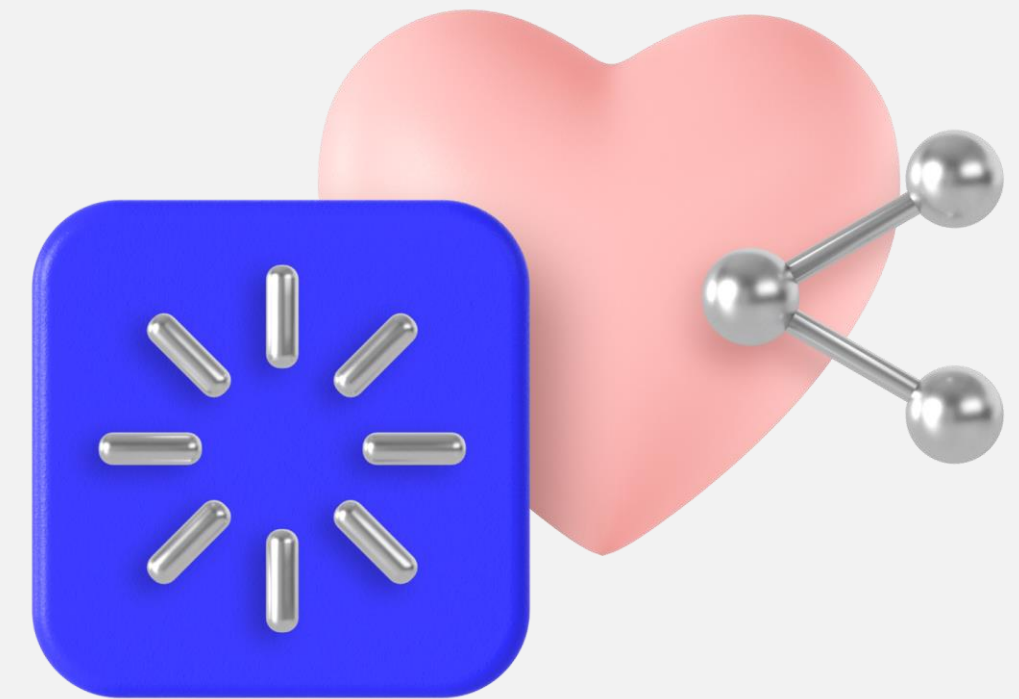


r -way replication: r replicas in r nodes
3-way most commonly used

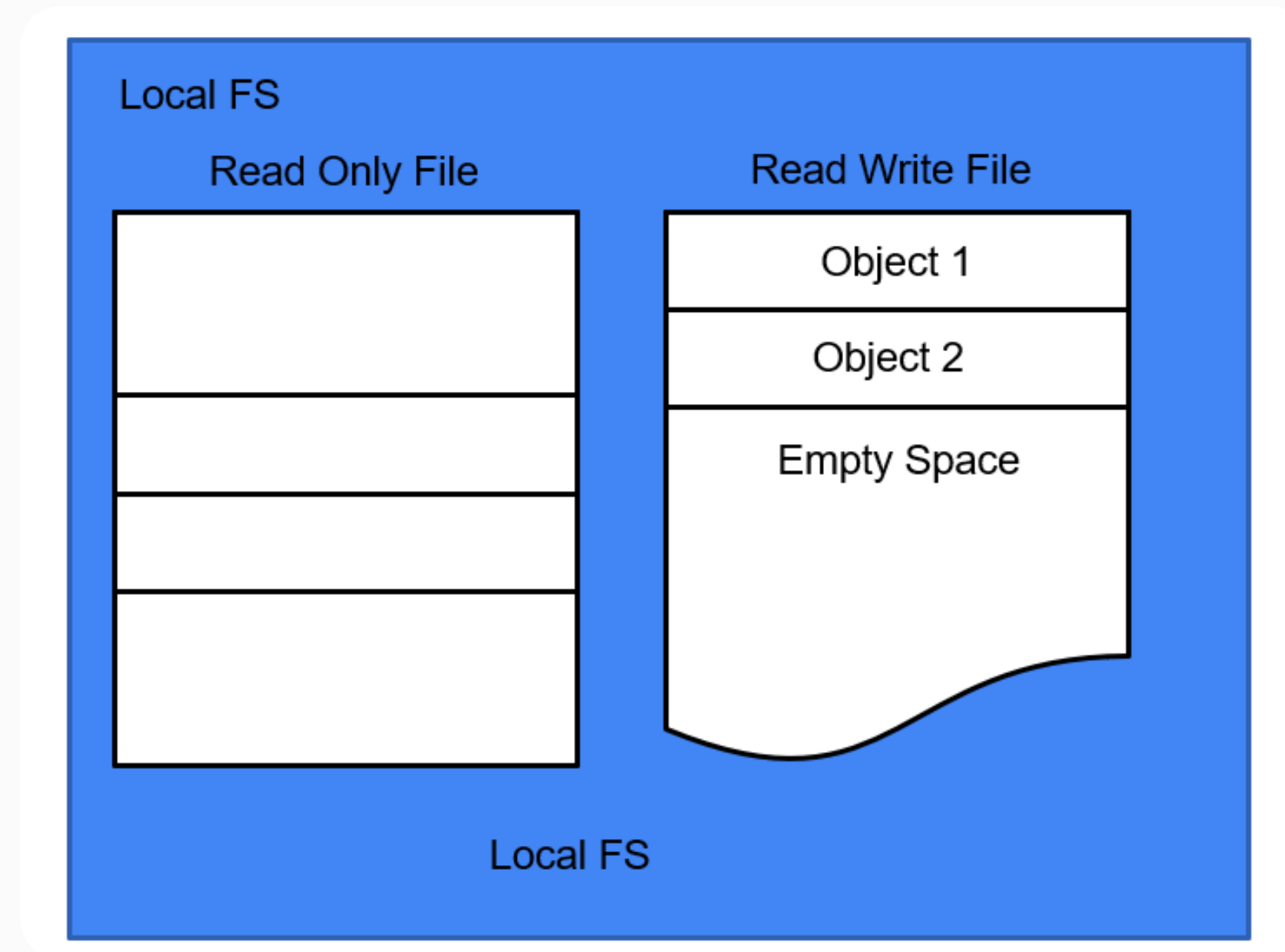


(n, k) erasure code: $(4, 2)$ in the example
 k data blocks $\rightarrow n - k$ parity blocks
 k data or parity blocks $\rightarrow$ original data

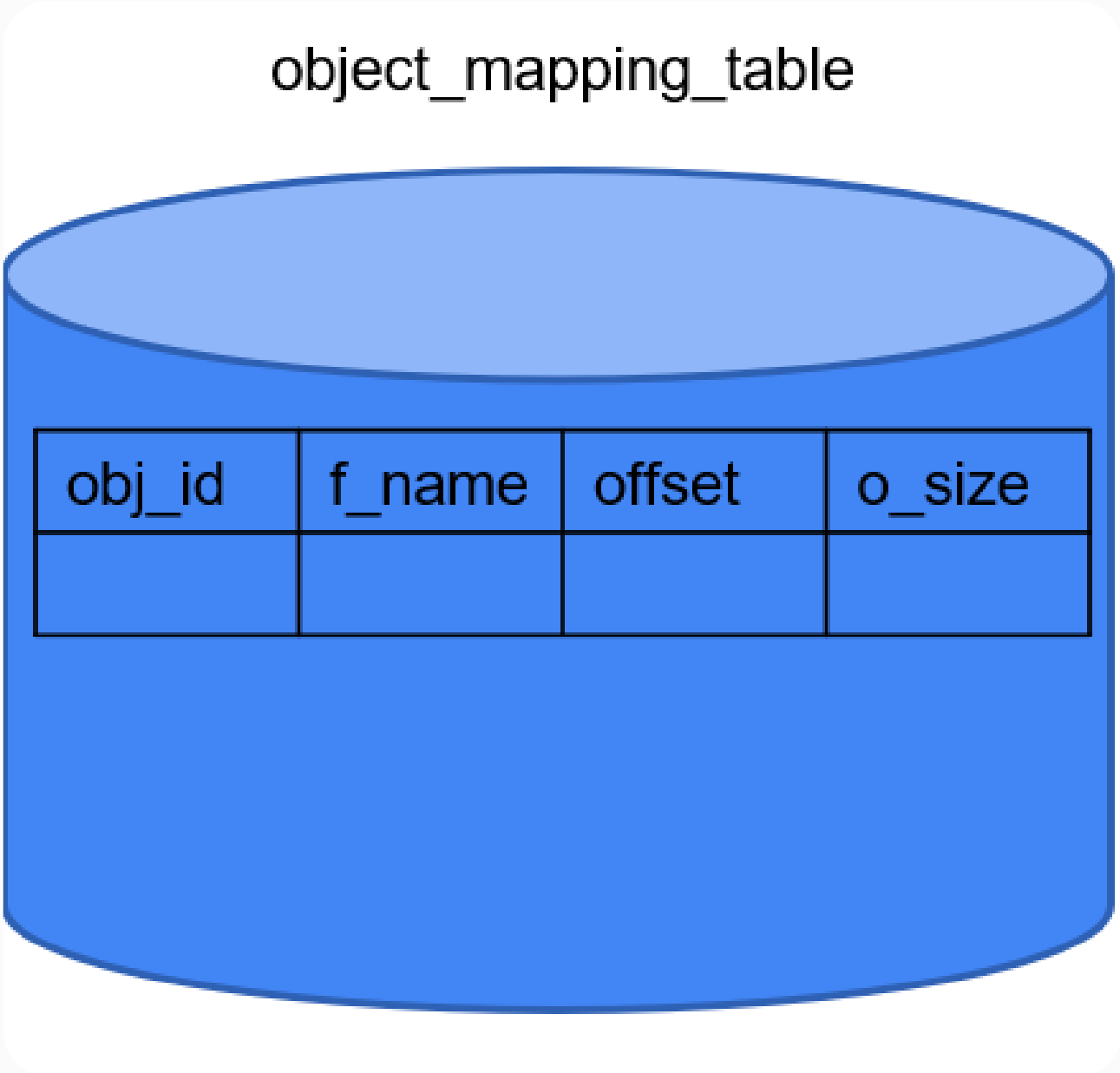
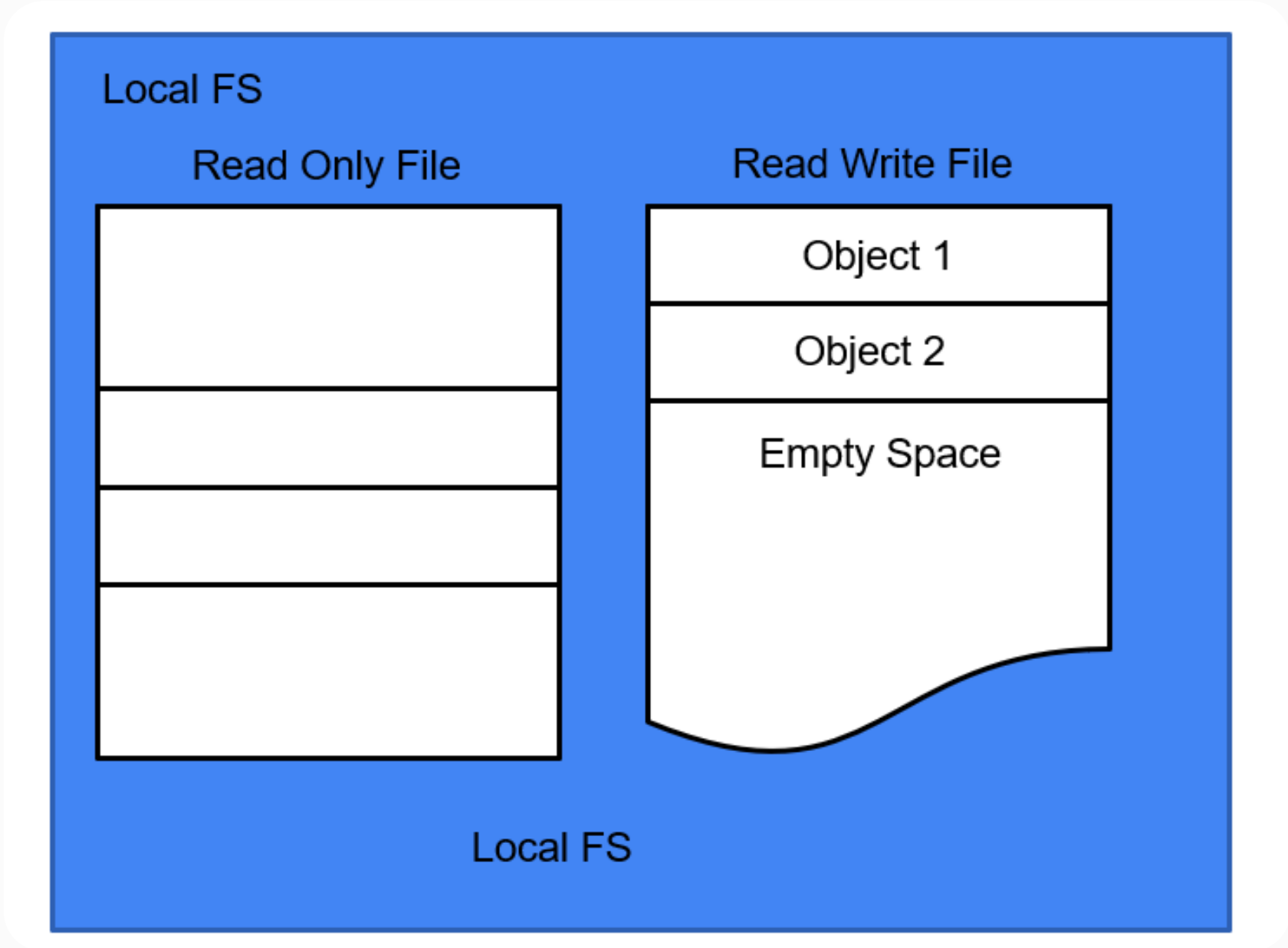
Small Files Problem



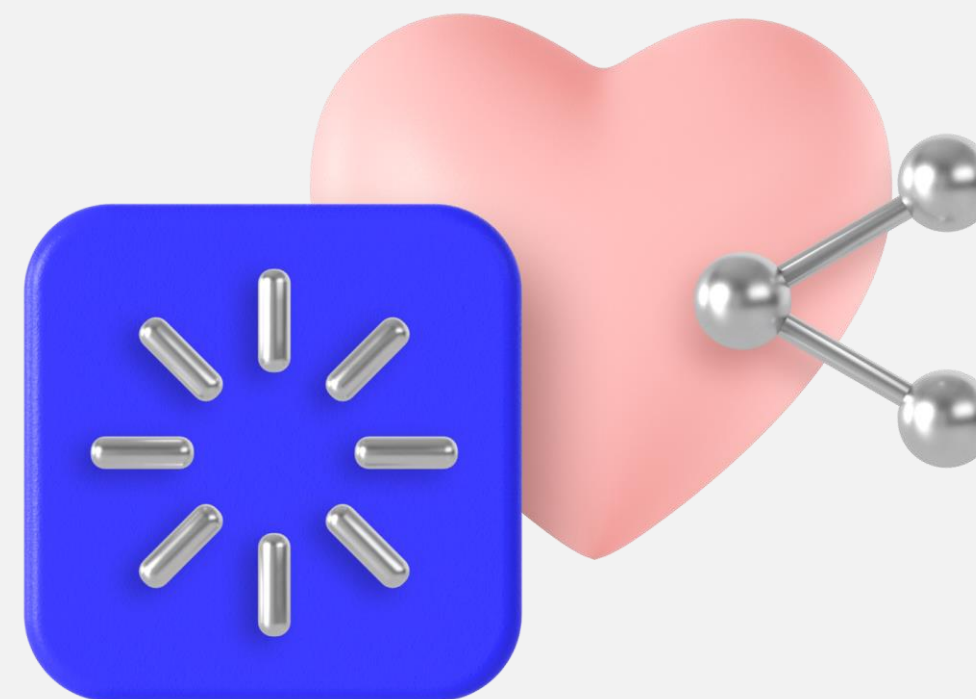
Virtual cluster map



Virtual cluster map



Примеры



Примеры

- 1 Easy Yandex S3
- 2 Amazon S3
- 3 Google Cloud Storage
- 4 Microsoft Azure

Итоги

- ❶ Рассмотрели основные типы и основные характеристики объектных хранилищ
- ❷ Познакомились с преимуществами объектных хранилищ